OS Orange Problem 2 (17/04/2026)

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<u>Analysis Questions' Answers</u>:

Q5.1: A branch in Git is just a file in .git/refs/heads/ containing a commit hash. Creating a branch is creating a file. Given this, how would you implement pes checkout <branch> — what files need to change in .pes/, and what must happen to the working directory? What makes this operation complex?

A:

To implement pes checkout <branch>:

1. Check if .pes/refs/heads/<branch> exists.
2. Read the commit hash from that branch file.
3. Read that commit and its tree from the object store.
4. Update working directory files to match that tree.
5. Update .pes/HEAD to point to ref: refs/heads/<branch>.
6. Update .pes/index so staged state matches checked out files.

Why this is complex:

1. You must safely replace files and folders.
2. You must avoid deleting user untracked files.
3. You must stop when local changes will be lost.

Q5.2: When switching branches, the working directory must be updated to match the target branch's tree. If the user has uncommitted changes to a tracked file, and that file differs between branches, checkout must refuse. Describe how you would detect this "dirty working directory" conflict using only the index and the object store.

A:

To detect dirty working directory conflict using index + object store:

1. For each tracked path in index, compare working file metadata with index metadata.
2. If different, re-hash working file and compare with index hash.
3. If hash differs, file is locally changed.
4. Read target branch tree and get target hash for same path.
5. If local file is changed and target hash is also different, checkout must stop.

This prevents overwriting user work.

Q5.3: "Detached HEAD" means HEAD contains a commit hash directly instead of a branch reference. What happens if you make commits in this state? How could a user recover those commits?

A:

If you commit in this state:

1. New commits are created normally.
2. But no branch name moves to those commits.
3. Later, these commits can become hard to find.

How to recover:

1. Create a branch at that commit hash as soon as possible.
2. If lost, find commit hash from history/reflog-like data and create a branch there.

Q6.1: Over time, the object store accumulates unreachable objects — blobs, trees, or commits that no branch points to (directly or transitively). Describe an algorithm to find and delete these objects. What data structure would you use to track "reachable" hashes efficiently? For a repository with 100,000 commits and 50 branches, estimate how many objects you'd need to visit.

A:

Algorithm to delete unreachable objects:

1. Start from all branch tips in .pes/refs/heads/.
2. Mark all reachable commits by following parent links.
3. For each reachable commit, mark its tree.
4. Walk tree entries recursively and mark subtree and blob hashes.
5. Scan .pes/objects and delete anything not marked.

Best data structure:

1. Use a hash set for marked object hashes.
2. It gives fast lookup while scanning objects.

For 100,000 commits and 50 branches:

1. You usually visit around 100,000 unique commits, not 5,000,000.
2. Then also visit related tree and blob objects.
3. Total objects can be much higher depending on file changes.

Q6.2: Why is it dangerous to run garbage collection concurrently with a commit operation? Describe a race condition where GC could delete an object that a concurrent commit is about to reference. How does Git's real GC avoid this?

A:

Running GC while commit is happening is risky because of race conditions.

Example race:

1. Commit process writes new blob/tree objects first.
2. Before branch ref is updated, GC checks reachable objects.
3. GC does not see those new objects from refs, so it deletes them.
4. Commit then updates branch to point to commit using deleted objects.
5. Repository gets broken because referenced objects are missing.

How real Git avoids this:

1. Uses locks and safe ordering.
2. Uses grace periods and reflog windows.
3. Runs GC in a way that avoids deleting recently created objects too soon.

Screenshots:

1A: `./test_objects` output showing all tests passing

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ export PES_AUTHOR="Shreyas P Kulkarni <PES1UG24CS444>"
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function 'object_write':
object.c:141:45: warning: '/.tmp-' directive output may be truncated writing 6 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
  141 |     snprintf(tmp_path, sizeof(tmp_path), "%s/.tmp-%ld-%d", shard_dir, (long)getpid(), rand());
      |                                             ^~~~~~
object.c:141:42: note: directive argument in the range [-2147483648, 2147483647]
  141 |     snprintf(tmp_path, sizeof(tmp_path), "%s/.tmp-%ld-%d", shard_dir, (long)getpid(), rand());
      |                                            ^~~~~~~~~~~~~~~
In file included from /usr/include/stdio.h:894,
                 from pes.h:8,
                 from object.c:10:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:71:10: note: '__builtin___snprintf_chk' output between 10 and 541 bytes into a destination of size 512
   71 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   72 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   73 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B: `find .pes/objects -type f` showing sharded directory structure

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ find .pes/objects -type f
.pes/objects/d1/b66183e2de3c4922aab1664b69bd4e882cd74587728113d3dd6baf1a1268ad
.pes/objects/2e/60c910e728d21156fb1b2baedbb0a36943608cb08af2b2371e085ccb58356b
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/14/3da3c3962c561d4305230fdd942e635846baf739c5d6c56b8c60946e80a09a
.pes/objects/d8/acd24d2a691088a7ad3bc5cda00b35eee61a5a78e6bb8f8ebcc3351df387c0
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/0f/8c952657aed100bbef7462c5d4a510fd210f25a011ff8fb176afbb63c91d87
.pes/objects/06/e1b4d91ebc08edabcae96fd540e2e56d52213c0f7700999975676f8cbc6378
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$
```

2A: `./test_tree` output showing all tests passing

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B: `xxd` of a raw tree object (first 20 lines)

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ xxd .pes/objects/d1/b66183e2de3c4922aab1664b69bd4e882cd74587728113d3dd6baf1a1268ad | head -20
00000000: 626c 6f62 2035 0077 6861 740a            blob 5.what.
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ find .pes/objects -type f
.pes/objects/d1/b66183e2de3c4922aab1664b69bd4e882cd74587728113d3dd6baf1a1268ad
.pes/objects/2e/60c910e728d21156fb1b2baedbb0a36943608cb08af2b2371e085ccb58356b
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
```

3A: `pes init` → `pes add` → `pes status` sequence (top part)

3B: `cat .pes/index` showing the text-format index

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index    # Human-readable text format
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  .gitignore
  untracked:  index.c
  untracked:  hello.txt
  untracked:  tree.h
  untracked:  tree.c
  untracked:  object.c
  untracked:  pes.h
  untracked:  test_tree.c
  untracked:  index.h
  untracked:  test_sequence.sh
  untracked:  what.txt
  untracked:  bye.txt
  untracked:  README.md
  untracked:  commit.c
  untracked:  test_tree
  untracked:  test_objects
  untracked:  Makefile
  untracked:  pes.c
  untracked:  commit.h
  untracked:  test_objects.c

100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776431383 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776431383 6 file2.txt
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776431383 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776431383 6 file2.txt
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$
```

4A: `pes log` output with three commits

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: ef8a2a9c35b0... Initial commit
Committed: be208c78bfce... Add world
Committed: 3c8be091f7f8... Add farewell
commit 3c8be091f7f8e66af9f4236967f2ef5eb49bc3f76b9953f9358a28d4cfa0c625
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431468

    Add farewell

commit be208c78bfceaa9a54a683e8b58034175967437efbe0a0eb0a7eb497e547d59d
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431468

    Add world

commit ef8a2a9c35b041ce6f8f6cd519ccdffe5aaf09cea851991336530d8c41cee6cb
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431468

    Initial commit

seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$
```

4B: `find .pes -type f \| sort` showing object growth

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/3c/8be091f7f8e66af9f4236967f2ef5eb49bc3f76b9953f9358a28d4cfa0c625
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/be/208c78bfceaa9a54a683e8b58034175967437efbe0a0eb0a7eb497e547d59d
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/ef/8a2a9c35b041ce6f8f6cd519ccdffe5aaf09cea851991336530d8c41cee6cb
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

4C: `cat .pes/refs/heads/main` and `cat .pes/HEAD`

```
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ cat .pes/refs/heads/main
3c8be091f7f8e66af9f4236967f2ef5eb49bc3f76b9953f9358a28d4cfa0c625
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$
```

Final: Integration Test

```
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 80cf8126f53b... Initial commit

Log after first commit:
commit 80cf8126f53b3eb61d304df52fbf564b4d65e5a7afd375c45d03dbbf62705dd0
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431725

    Initial commit


--- Second Commit ---
Committed: 1eb0fc4088d0... Update file.txt

--- Third Commit ---
Committed: 2ef60046cf6a... Add farewell

--- Full History ---
commit 2ef60046cf6a7fdec1d534dc20edc5147cf37f8d7b72df23ef509c9f8cb4dba6
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431726

    Add farewell
```

```
--- Third Commit ---
Committed: 2ef60046cf6a... Add farewell

--- Full History ---
commit 2ef60046cf6a7fdec1d534dc20edc5147cf37f8d7b72df23ef509c9f8cb4dba6
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431726

    Add farewell

commit 1eb0fc4088d0ed9c77fb9374482a0a1ae1ab241afb6114e96bac61968c1343df
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431726

    Update file.txt

commit 80cf8126f53b3eb61d304df52fbf564b4d65e5a7afd375c45d03dbbf62705dd0
Author: Shreyas P Kulkarni <PES1UG24CS444>
Date:   1776431725

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
2ef60046cf6a7fdec1d534dc20edc5147cf37f8d7b72df23ef509c9f8cb4dba6

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/1e/b0fc4088d0ed9c77fb9374482a0a1ae1ab241afb6114e96bac61968c1343df
.pes/objects/2e/f60046cf6a7fdec1d534dc20edc5147cf37f8d7b72df23ef509c9f8cb4dba6
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/80/cf8126f53b3eb61d304df52fbf564b4d65e5a7afd375c45d03dbbf62705dd0
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
seed@seedvm2006:~/Documents/PES1UG24CS444-pes-vcs$
```